THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 8: **Fractions in Disguise** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The CI on a sum for 2 years at 25% p.a. is ₹ 3600. The sum (in ₹) is: 1

(a) 6400 **(b)** 12800 **(c)** 5760 **(d)** 7040

2. A trader marks his goods 60% above cost and allows a discount of 25%. His gain percent is: 1

(a) 18% **(b)** 35% **(c)** 20% **(d)** 22%

3. If the difference between the CI and the SI on a sum for 2 years at 5% is ₹ 10, the sum (in ₹) is: 1

(a) 4000 **(b)** 8000 **(c)** 4400 **(d)** 3600

4. A sum triples in 10 years at simple interest. The rate of interest per annum is: 1

(a) 20% **(b)** 18% **(c)** 22% **(d)** 40%

5. Two articles are sold at ₹ 1500 each, one at a gain of 30% and the other at a loss of 30%. The overall result is a loss of (in %): 1

(a) 9% **(b)** 30% **(c)** 15% **(d)** 0%

6. The population of a town is 80000 and it grows at 25% per annum. After 2 years it will be: 1

(a) 125000 **(b)** 137500 **(c)** 112500 **(d)** 120000

7. A machine worth ₹ 40000 depreciates at 5% per annum. Its value after 2 years (in ₹) is: 1

(a) 32490 **(b)** 36000 **(c)** 39710 **(d)** 36100

8. A price is first increased by 50% and then decreased by 40%. The net change (as percent, + for increase, − for decrease) is: 1

(a) −11% **(b)** −9% **(c)** +10% **(d)** −10%

9. **Assertion (A):** The simple interest on ₹2,000 at 5% per annum for 3 years is ₹300. 1

Reason (R): Simple interest $= \dfrac{P \times R \times T}{100}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The compound interest on ₹5,000 at 10% per annum for 2 years is ₹1,050. 1

Reason (R): For the first year, simple interest and compound interest on the same principal at the same rate are equal.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Rahul invests ₹ 1600 at 25% p.a. compounded annually. Find the amount after 2 years and write it as a fraction of the principal. 2

12. A number is increased by 60% and the result is decreased by 50%. Is the final number more or less than the original? By what percent? 2

13. The cost of an item is ₹ 2000. It is marked 60% above the cost price and sold at 25% discount. Find the selling price. 2

14. Rahul invests ₹ 1000 at 30% p.a. compounded annually. Find the amount after 2 years and write it as a fraction of the principal. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A merchant sells two items at ₹ 990 each, one at a gain of 10% and the other at a loss of 10%. Find the overall gain or loss in rupees. 3

16. The value of a car depreciates by 25% every year. If its present value is ₹ 160000, find its value after 3 years. 3

17. A shopkeeper buys an article at ₹ 1500, marks it up by 20% and gives a discount of 25%. Find his gain or loss percent. 3

OR

A merchant sells two items at ₹ 1200 each, one at a gain of 20% and the other at a loss of 20%. Find the overall gain or loss in rupees. 3

18. The value of a car depreciates by 20% every year. If its present value is ₹ 625000, find its value after 3 years. 3

OR

A shopkeeper buys an article at ₹ 900, marks it up by 80% and gives a discount of 40%. Find his gain or loss percent. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Farmer Gurmeet takes a loan of ₹8,000 at 10% per annum. 5

(i) Find the simple interest for 1 year.
(ii) Find the simple interest for 3 years.
(iii) Find the amount payable after 2 years at compound interest.
(iv) Find the difference between the compound interest and the simple interest for 2 years.

20. A sum of ₹ 30000 is invested for 3 years at 10% p.a. compounded annually. Find the interest for each of the three years separately and verify that their total is the compound interest. 5

OR

A shopkeeper marks an article 100% above its cost price. During a sale he gives a discount of 40% and still earns a profit of ₹ 400. Find the cost price. 5

END OF QUESTION PAPER — Total: 40 Marks